JAMES MITCHELL

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 7 years of experience building and deploying production ML systems across finance and healthcare sectors. I've spent the last three years leading technical ML initiatives at fast-growing startups, mentoring junior engineers, and shipping models that directly impact user outcomes. My focus is on practical deep learning applications rather than theoretical research, with particular expertise in computer vision and time-series forecasting. I'm based in Oxford and looking for roles where I can take on greater architectural responsibility.

WORK EXPERIENCE

Senior Machine Learning Engineer

DataFlow Systems Ltd, oxford.dataflow.io

Oxford, United Kingdom

March 2022 - Present

Led the design and implementation of a real-time anomaly detection pipeline processing 50+ million transactions daily using LSTM networks and isolation forests, reducing false positives by 35%. Mentored a team of three junior ML engineers on best practices in model validation, feature engineering, and production deployment strategies. Architected a feature store using Apache Feast and PostgreSQL that reduced model training time from 4 hours to 12 minutes across 15 active models. Established ML monitoring and observability practices using Prometheus and custom dashboards, catching model drift issues before they impacted business metrics.

Machine Learning Engineer

HealthTech Analytics, healthtech.co

Cambridge, United Kingdom

June 2019 - February 2022

Developed and deployed a chest X-ray classification model using ResNet-50 and transfer learning, achieving 94% sensitivity on rare pneumonia cases and integrating into the diagnostic workflow of three NHS trusts. Built an automated data pipeline using Airflow that reduced manual data preparation time from two weeks to four hours, handling DICOM image standardization and anonymization. Implemented a Bayesian hyperparameter optimization framework using Optuna that improved model performance across six concurrent projects while reducing GPU compute costs by 28%. Collaborated with clinical teams to gather requirements and validate model predictions on real patient data, ensuring medical accuracy and regulatory compliance.

Data Scientist

Thomson Financial Services, thomson-fs.net

London, United Kingdom

July 2017 - May 2019

Created machine learning models for algorithmic trading, including gradient boosting models for market direction prediction that achieved 58% accuracy on out-of-sample test sets and generated consistent returns during backtesting. Developed a recommendation system using collaborative filtering and matrix factorization to suggest financial products to retail clients, increasing conversion rates by 12%. Processed and analyzed large datasets

using PySpark and SQL, building ETL pipelines that prepared market data and alternative data sources for model consumption. Documented model implementations in technical reports and presented findings to non-technical stakeholders including product managers and compliance officers.

EDUCATION

University of Oxford

MSc Computer Science (Machine Learning specialization)

Oxford, United Kingdom

2015 - 2017

University of Manchester

BSc Mathematics and Computer Science

Manchester, United Kingdom

2011 - 2015

TECHNICAL SKILLS

Machine Learning: Deep Learning, Computer Vision, Time-Series Forecasting, Anomaly Detection, Classification, Regression, Feature Engineering, Model Validation

Programming Languages: Python, SQL, Scala, R

Deep Learning Frameworks: TensorFlow, PyTorch, Keras

ML Tools and Libraries: scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Optuna, Hyperopt

Data Processing: Apache Spark, Apache Airflow, Pandas, SQL

Databases: PostgreSQL, MongoDB, MySQL, HBase

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform (Vertex AI, BigQuery), Azure ML

DevOps and Deployment: Docker, Kubernetes, Git, Jenkins, MLflow, Weights and Biases

Monitoring and Observability: Prometheus, Grafana, custom logging solutions

GITHUB REPOSITORIES

github.com/jamesmitchell/time-series-anomaly-detection

Comprehensive implementation of LSTM and Isolation Forest anomaly detection with real financial transaction data and visualization tools

github.com/jamesmitchell/medical-image-classifier

ResNet-50 based chest X-ray classification model with data augmentation pipeline and Grad-CAM visualization for model interpretability

github.com/jamesmitchell/feature-store-framework

Production-ready feature store implementation using Apache Feast and PostgreSQL with examples and deployment documentation

github.com/jamesmitchell/hyperparameter-optimization-benchmark

Benchmark study comparing Optuna, Hyperopt, and grid search across multiple datasets with performance analysis and recommendations